


# Name: Abdul Samad

# Program: BSSE Group A

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# Lab Task 2 - Report

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*Scenario: Online Food Delivery System*

### 1. Problem Statement

Customers browse restaurants and place orders. Restaurants receive orders and update order status. Delivery riders receive delivery requests and update delivery status. A payment service processes online payments. An administrator manages users, restaurants, and complaints.

## 2. Extracted Requirements

- R1: Customers must be able to browse restaurants/menus and place orders.
- R2: Customers must be able to pay for orders online.
- R3: Restaurants must be able to receive orders and update order status.
- R4: Delivery riders must be able to receive delivery requests and update delivery status.
- R5: The system must communicate with a payment service to process payments.
- R6: An administrator must be able to manage users, restaurants, and complaints.

## 3. Identify External Entities

The following entities are outside the boundary of the Online Food Delivery System but interact with it:

| **External Entity** | **Why External?** | **Major Interaction** |
|---|---|---|
| Customer | A person using the system, not an internal software component. | Requests restaurant/menu information, places orders, makes payments, receives order status. |
| Restaurant | An independent business external to the software system. | Receives new orders; sends order status updates. |
| Delivery Rider | A person outside the software system who fulfils deliveries. | Receives delivery requests; sends delivery status updates. |
| Payment Service | A separate external paymentprocessing service. | Receives payment requests; sends payment confirmation. |
| Administrator | A person managing the platform, not an internal component. | Manages users, restaurants, and complaints. |

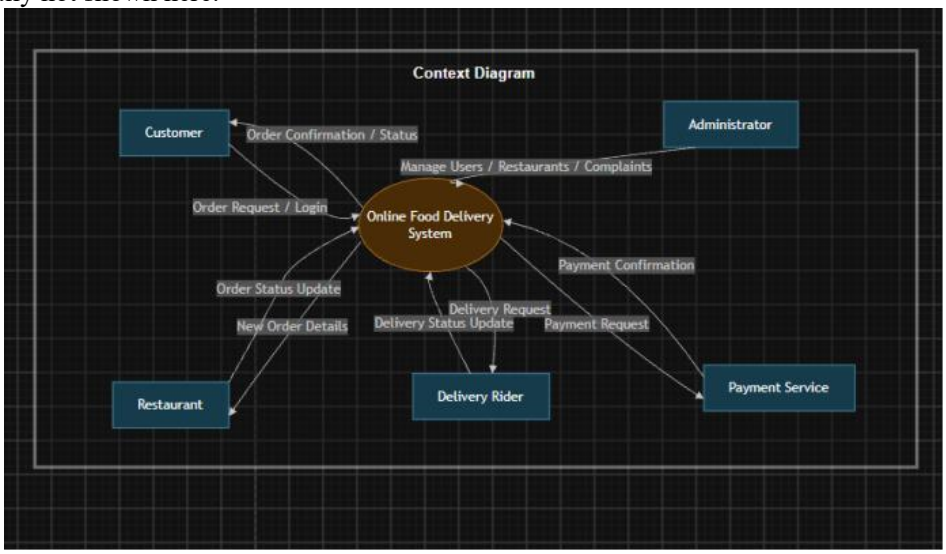
# 4. Identify Major Information Flow

| **External Entity** | **Information Sent to System** | **Information Received** | **Related Requirements** |
|---|---|---|---|
| Customer | Browse request, order placement request, payment info | Restaurant/menu list, order confirmation, order status updates | R1, R2 |
| Restaurant | Order status update | New order details | R3 |
| Delivery Rider | Delivery status update | Delivery request | R4 |
| Payment Service | Payment confirmation | Payment request | R5 |

| | | | |
|---|---|---|---|
| Administrator | Manage users / restaurants / complaints | Management confirmation / records | R6 |

# 5. Context Diagram

The context diagram below treats the entire Online Food Delivery System as one unit. Customer, Restaurant, Delivery Rider, Payment Service, and Administrator remain outside the system boundary. Internal modules are intentionally not shown here.




## Explanation

- The central node represents the complete software system.
- Customer, Restaurant, Delivery Rider, Payment Service, and Administrator are external entities.
- Flows are labeled with meaningful information (e.g., "Order Request / Login") rather than generic terms such as "data".
- The diagram does not show databases, screens, classes, APIs, or internal modules.
- Payment Service is external because payment processing is performed by a separate service, not by the ordering platform itself.

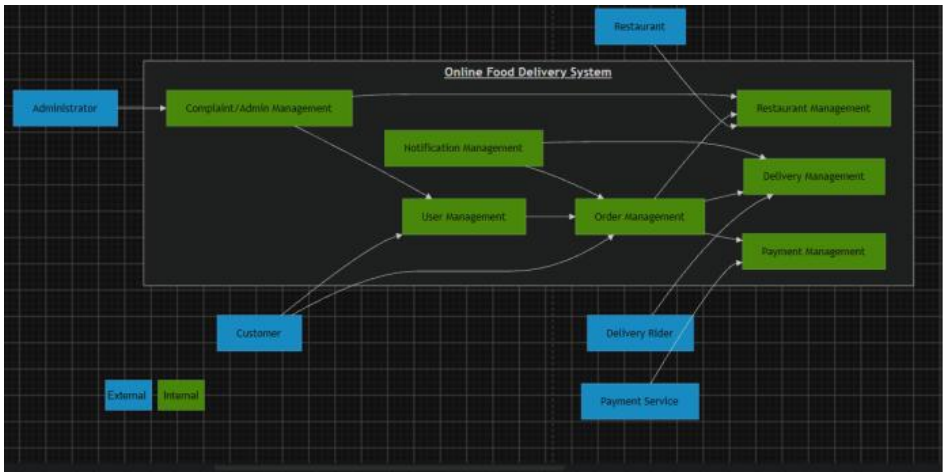
# 6. Major Internal Modules

| **Internal Module** | **Responsibility** |
|---|---|
| User Management | Manages customer, restaurant, and rider accounts and profiles. |
| Restaurant Management | Maintains restaurant listings, menus, and availability. |
| Order Management | Processes order placement and coordinates order status across modules. |
| Delivery Management | Assigns delivery requests to riders and tracks delivery status. |

| | |
|---|---|
| Payment Management | Handles payment requests and communicates with the external Payment Service. |
| Notification Management | Sends status updates and alerts to customers, restaurants, and riders. |
| Complaint/Admin Management | Allows the administrator to manage users, restaurants, and complaints. |

# 7. System Diagram

The system diagram opens the system boundary and shows the major internal modules and their relationships. It remains high-level and does not describe source code classes or individual database fields.




## Explanation

- Order Management is the central coordinating module, connecting to Restaurant Management, Delivery Management, and Payment Management.
- Notification Management supports Order Management and Delivery Management by sending status updates.
- User Management handles account-related requests from Customers and feeds into Order Management.
- Complaint/Admin Management gives the Administrator control over User Management and Restaurant Management (e.g approving/removing restaurants, resolving complaints).
- Restaurant Management, Delivery Management, and Payment Management each interface directly with their corresponding external entity (Restaurant, Delivery Rider, Payment Service).

# 8. Context Diagram vs. System Diagram

| **Aspect** | **Context Diagram** | **System Diagram** |
|---|---|---|
| Perspective | External / boundary view | Internal high-level view |
| System representation | One complete system | Major internal modules |
| External entities | Shown | Shown where useful for interaction |

| | | |
|---|---|---|
| | | context |
| Internal modules | Not shown | Shown |
| Main purpose | Define boundary and external interactions | Explain major internal organization |
| Level of detail | Very high level | More detailed, but still conceptual |

# 9. Requirement to Diagram Traceability

| **Requirement** | **Context Diagram Evidence** | **System Diagram Evidence** |
|---|---|---|
| R1 Browse & order | Customer <-> System (Order Request / Login) | User Management -> Order Management |
| R2 Online payment | System <-> Payment Service | Order Management -> Payment Management |
| R3 Restaurant order/status | System <-> Restaurant | Order Management <-> Restaurant Management |
| R4 Delivery request/status | System <-> Delivery Rider | Order Management / Notification Management <-> Delivery Management |
| R5 Payment processing | System <-> Payment Service | Payment Management <-> Payment Service |
| R6 Admin manages users/restaurants/complaints | Administrator <-> System | Complaint/Admin Management <-> User Management, Restaurant Management |

# 10. Ambiguities and Assumptions

## Ambiguities

- The problem statement does not specify whether Notification Management is a requirement or an implementation choice, it was added as a supporting module for order status updates.
- The scope of "complaints" is not defined , it is unclear whether complaints can be raised by customers, restaurants, or riders, or all three.

## Assumptions

- Payment Service is assumed to be a separate external system.
- Restaurants are assumed to have some interface (app/portal) to receive orders and update status, since the requirement states they "receive orders and update order status".
- Administrator is assumed to be a single internal role with access to Complaint/Admin Management, rather than an external organization.